Dual Mode Logic Carry Look Ahead Adder

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***Abstract*- Carry Look-ahead Adder (CLA) is implemented by using Dual Mode Logic (DML) topologies. DML logic switches between the static and dynamic mode of operations. In dynamic mode achieves higher performance with increase in power consumption and in static mode, DML logic achieves low power dissipation albeit with reduced performance. This feature allowed implementing CLA by selection of carry path based on input vectors. A 4-bit CLA was designed in 45nm TSMC technology using Cadence Virtuoso Design. Simulation results showed gain in speed albeit with increase in power and area when compared to the conventional CMOS logic.**

Keywords–Carry Look Ahead adder, CMOS, Dual Mode Logic.

I. INTRODUCTION

Mostly every arithmetic operation requires adders [1]. Addition operations can be main instruction performed while processing [2]. Many adders' architectures are designed and proposed for speed improvement and power reduction [3]. When designing faster adders, it is essential to improve delay due to the many gates in the carry path from the Least Significant Bit to the Most Significant Bit. Among them Carry Look-ahead Adder (CLA) [4] was an attractive adder with reduced delay at the cost of complex hardware. The CLA design can be obtained by transformation of the ripple carry design into a design in which the carry logic over groups of bits of the adder is reduced to two-level logic.

In this paper we present a high performance and power efficient CLA implementation. According to the proposed logic critical carry paths of the CLA [4] are dynamically identified based on the input vectors. Carry paths are implemented by using Dual Mode Logic (DML) [5]. Dual Mode Logic switches between static and dynamic modes of operation. In the static mode, the DML gates feature very low power dissipation with moderate performance and in the dynamic mode they achieve high performance with higher power dissipation. This feature of DML offers ability to control performance in the proposed CLA. Delay and power consumption are the parameters compared with the conventional CMOS [6] logic CLA adder and the DML logic CLA adder.

II. DUAL MODE LOGIC

With advancements in technology and the expansion of mobile applications the demand for high processing capabilities increased. The DML logic was able to allow switching between static and dynamic modes of operation. The basic architecture of DML logic was composed of a standard CMOS gate and an additional transistor M1, whose gate is connected to a global clock signal. DML allow operations between static and dynamic mode. To operate the gate in dynamic mode, the clock is assigned allowing two distinct phases: precharge and evaluation. During the precharge phase, the output is charged to high/low, depending on the topology of the DML gate. In the consequent evaluation phase, the output is evaluated according to the values at the gate inputs. Dual Mode Logic (DML) is classified into two topologies: Type-A and Type-B topology.

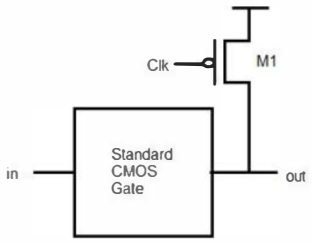


Figure 1. DML Type –A topology

Fig. 4 shows the DML Type –A topology. In this topology standard CMOS Gate is connected by an additional PMOS transistor at the output. The Clk signal drives the additional M1 transistor to switches between Static and Dynamic modes. In Static Mode the transistor is in off state i.e. the Clk signal is high. When the transistor is on, the Clk signal will be asymmetric clock, allowing two different phases: precharge and evaluation. In precharge phase, the output is high and in dynamic phase the output is evaluated based on the inputs of the CMOS network. DML type –B topology is shown in Fig. 5, in the type –B topology, a NMOS transistor is glued to the standard CMOS network. A Clock (Clk) signal drives the NMOS transistor as explained in Type–A topology.

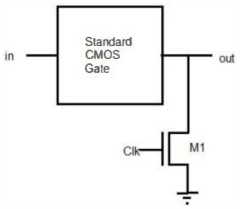


Figure 2. DML Type –B topology

When Clk signal is high the M1 transistor makes the output low in precharge phase and the outputs are evaluated in the evaluation phase of Dynamic phase. The standard problems like glitching will arise after precharge ends and until the evaluation data ripples through the chain

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[7]. The minimum sized transistors are used as the precharge transistor to reduce the leakage currents during static operation and evaluation. DML logic posses smaller than or equal delay to its CMOS counterparts. Where the precharge phase and power consumption are not constraining, DML logic is preferable.

III. CLA ADDER IMPLEMENTATION USING DML

The ripple carry adder is simple in concept but has a long circuit delay due to many gates in the carry path from the significant bit to the most significant bit. In a computer design, delay is the one of the largest trend. The alternate design to reduce the delay is the carry look ahead Adder. Delay is reduced at the cost of complex design. Carry Lookahead Adder design can be obtained by transformation of the ripple carry design into a design in which carry the logic path is used separately. In CMOS implementation, the critical path will always be the longest carry route and is determined by the number of bits: 2 $\log_2(N)$-1. The carry path of the CLA to be dynamically chosen and accelerated by the operation of the DML gates of that path in the dynamic mode. Based on the inputs, carrypath is identified and the carrypath is accelerated. To accelerate the carrypath, decision logic is used, which identifies the carrypath and creates the clock signal for the DML gates in the required path.

A. Decision Logic

Decision logic is used for switching between dynamic to static modes of operation. Carrypath is dynamically operated only if the CLA inputs propagate carry. In the case of carryout propagation a simple 1-bit NOR operation is performed between the inputs which ensure the dynamic operation of the path. However, the NOR operation accelerate the 33% of the carrypath dynamically when the operation is not required, resulting the increase in power dissipation.

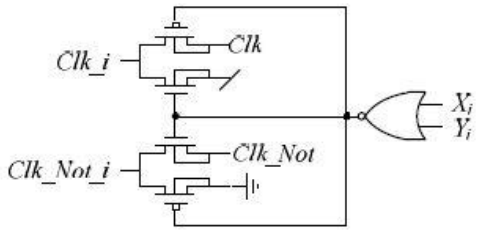


Figure 3. 1-bit Decision logic switch.

TABLE 1: DYNAMIC ACTIVATION AS A FUNCTION OF INPUT VECTORS

X_i	Y_i	C_{in}	C_{next}	NOR (X_i,Y_i)	False Dynamic activation
0	0	0	0	1	No
0	0	1	0	1	No
1	0	0	0	0	Yes
1	0	1	1	0	No
0	1	0	0	0	Yes
0	1	1	1	0	No
1	1	0	1	0	No
1	1	1	1	0	No

Two Low-Vth transistors are used to construct switch which is controlled by NOR gate output. The decision is based on input bits of X_i and Y_i and the outputs are connected to the pre-charge transistor of the DML topologies. Clk_i controls the DML Type A logic and Clk_Not_i controls the DML Type B logic. Where the dynamic operation is not required, the decision circuitry disables all the pre-charge transistors of DML logic on the carrypath, achieving static operation of DML gates. In two conditions, Carrypath is activated falsely as shown in the table. When NOR output is zero, carrypath is dynamically activated.

B. CLA Architecture

A new logic is constructed by separating the parts of the full adder not involving the carry propagation path from those containing the path. The first part of each full adder is a partial full adder (PFA) which provides the sum of the bits. There are two outputs P_i and G_i from each PFA and one input C_i, carry input, from the carry path to each PFA. These propagate and generate terms depend only on the input bits. These terms exactly corresponds to the half adder and are essential in controlling the values in the ripple carry path. The main core of the CLA adder is very similar to a conventional CMOS CLA adder which is composed of 2 blocks: A and B. The functionality of these blocks is

$$S_i = x_i \oplus y_i \oplus C_i \quad (1)$$

$$P_i = x_i \oplus y_i \quad (2)$$

$$G_i = x_i . y_i \quad (3)$$

$$C_{i+1} = G_i + P_i C_i \quad (4)$$

Where P_i and G_i are called propagate and generate terms. x_i and y_i are the two inputs given to the CLA Adder and C_i is the carry input at the ith stage. Carrypath of the CLA has 4 extra 1-bit decision circuits compared to the conventional CMOS implementation. Each circuit is used for the static or dynamic operation of a specific route. For example, if the inputs of X_i and Y_i are such that a carry is needed, then the $C_{A(i+1)}$ route would be dynamically operated. Fig. 5 shows the implementation of 4 bit CLA adder by using partial adder (A_i) and B block (carrypath). Fig. 4 shows inputs and outputs of B block. To implement higher order bit adder, multiple partial adders and B blocks are used. Every B block produces P_{bi} and G_{bi} , which are used to accelerate the signals of the CLA adder in the next level. If they are only four bits in the adder, then the logic circuit used for Ca1 can be used to generate Ca4 from those inputs. Fig. 6 shows the logic circuitry of B block.



Figure 4. Input and outputs of B Block

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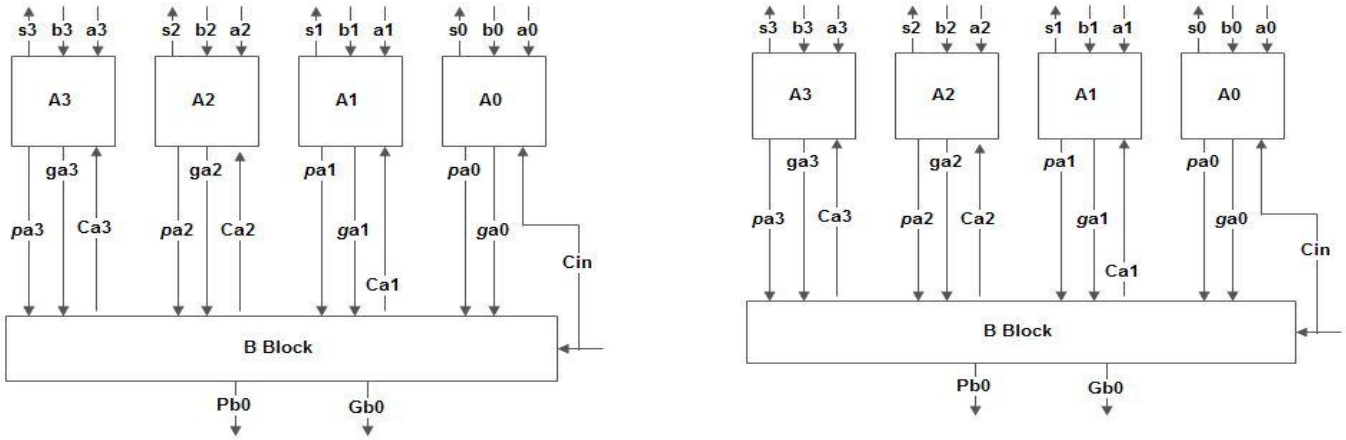


Figure 5. CLA Implementation

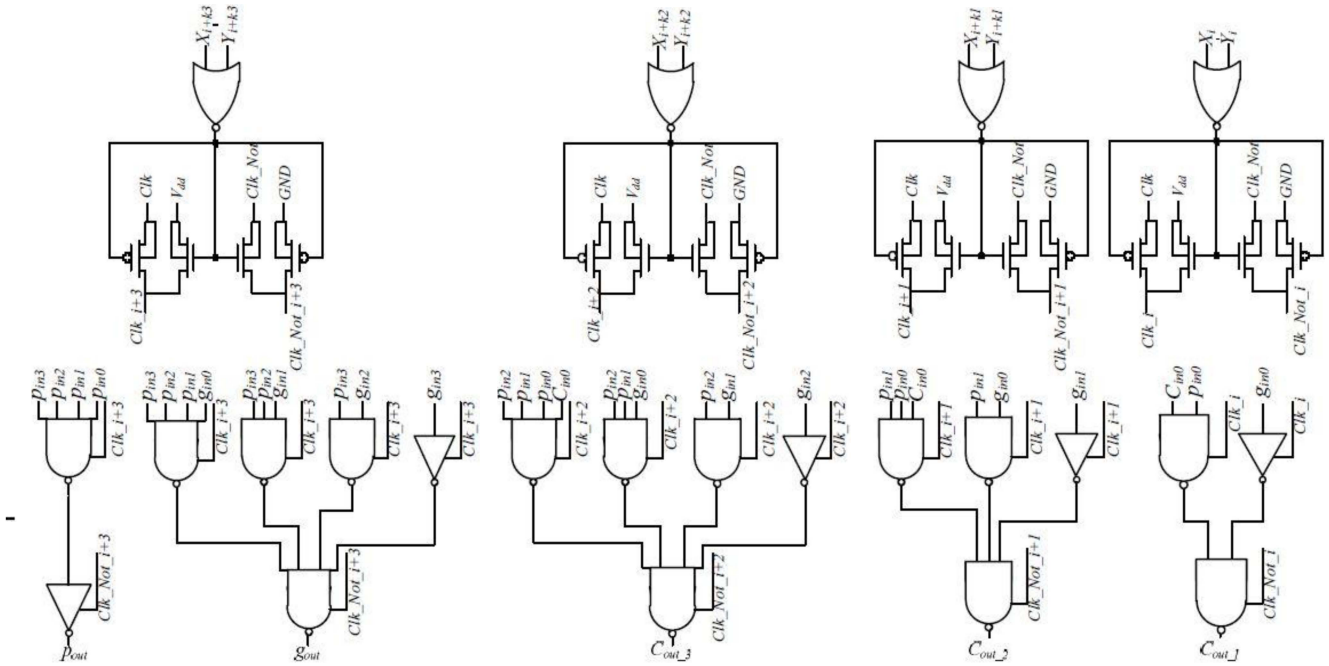


Figure 6. Structure of B Block

IV. Simulation Results

The DML logic is used to implement the CLA adder. The design is carried out in TSMC 045nm technology by using Cadence Virtuoso Design and the schematic is shown in Fig. 7. Comparison of delay, power consumption and Power-Delay Product is done between conventional CMOS logic CLA adder and the DML logic CLA adder for multiple bit adders and the results are given in the Table. II and Table III. The comparison of power-Delay product of multiple bit adders is graphically represented as shown in Fig. 8. Using Fig. 8 It has been found that the Power-Delay product is less for DML Logic CLA adder compared to the conventional CMOS CLA adder.

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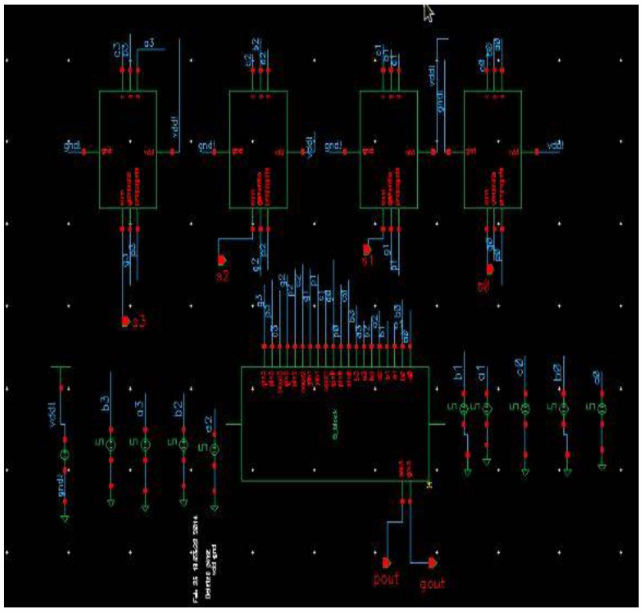


Figure 7. Implementation of DML logic CLA.

TABLE II. COMPARISON OF POWER AND DELAY FOR 4-BIT.

Design	Power (µW)	Delay (pS)	Power-Delay Product (pJ)
Conventional CMOS CLA adder	1.2	10540.6	0.123
DML logic CLA adder	54.5	100.9	0.005

TABLE III. COMPARISON OF POWER-DELAY PRODUCT FOR MULTIPLE BIT DATA.

Design	Power-Delay Product (pJ) - 8 bit	Power-Delay Product (pJ) -16 bit	Power-Delay Product (pJ) -32 bit
Conventional CMOS CLA adder	0.246	0.492	0.984
DML logic CLA adder	0.0108	0.0216	0.0432

From the simulated results Power-delay product (PDP) of the DML logic CLA adder is very efficient compared to the conventional CMOS CLA Adder. PDP is calculated for various bit vectors as shown in Table 3. Fig. 8 Shows the comparison of PDP of conventional CMOS and DML logic CLA adders.

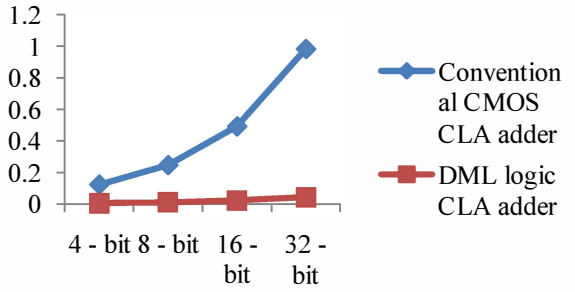


Figure 8. Power-Delay product Comparison of multiple bit adders.

V. CONCLUSION

The DML logic is used to reduce the delay of CLA adder compared to CMOS CLA adder where the power consumption is not a key parameter. DML logic is an efficient technique in reducing delay and showed in terms of picoseconds (pS). It gives the CMOS circuit designers another option in designing integrated circuits more efficiently. The entire designs are carried out in TSMC 045nm technology Cadence Virtuoso Design.

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